Planning, Learning and Intelligent Decision Making
Homework 2

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Exercise 1.

(a)

The states for this Markov Decision Problem are based on the position of the taxi driver and whether the passenger is or isn't in the car. Thus the state space $\mathcal{X}$ can be described as $\{(P,\ 1),\ (P,\ 2),\ (\neg P,\ 2),\ (P,\ 3),\ (\neg P,\ 3),\ (P,\ 4),\ (\neg P,\ 4)\}$, in which the first element states whether or not the passenger is in the taxi (P if the passenger is in the taxi, $\neg P$ otherwise) and the second where the taxi is on the grid. It is important to note that $(\neg P,\ 1)$ was not considered, since it is said that "as soon as the taxi driver stands on the cell marked with "P", you may consider that the passenger is in the car." The actions are the possible directions in which the car can move. So, $\mathcal{A} = \{U,\ D,\ L,\ R\}$, U stands for Up, D for down, L for left, and R for right.

(b)

The transition probability matrix for the action "down" is:

$$P_D = \begin{array}{c} \\ P1 \\ P2 \\ \neg P2 \\ P3 \\ \neg P3 \\ P4 \\ \neg P4 \end{array} \begin{array}{c} \begin{array}{ccccccc} P1 & P2 & \neg P2 & P3 & \neg P3 & P4 & \neg P4 \end{array} \\ \begin{bmatrix} 0.2 & 0 & 0 & 0.8 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 & 0 & 0.8 & 0 \\ 0 & 0 & 0.2 & 0 & 0 & 0 & 0.8 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Note: For simplicity, we adopted the convention: $(P,\ 2)$ as $P2$ and $(\neg P,\ 2)$ as $\neg P2$.

Concerning the cost, since in this problem there isn't a state that the taxi shouldn't go to, none of the actions should be penalized with the greatest cost, 1. However, the goal state should have the minimum cost, and all the other intermediate states should have an uniform cost. Therefore, let's consider 0.5 as the "intermediate" cost.

$$C = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}$$

(c)

Considering a policy π that always chooses the action "down", let's calculate its cost-to-go function using 0.9 as its discount, according to $J^\pi = (I - \gamma P_\pi)^{-1} C_\pi$.

$$
P_\pi = \begin{array}{r} \\ P1 \\ P2 \\ \neg P2 \\ P3 \\ \neg P3 \\ P4 \\ \neg P4 \end{array}
\begin{array}{c}
\begin{array}{ccccccc} P1 & P2 & \neg P2 & P3 & \neg P3 & P4 & \neg P4 \end{array} \\
\begin{bmatrix}
0.2 & 0 & 0 & 0.8 & 0 & 0 & 0 \\
0 & 0.2 & 0 & 0 & 0 & 0.8 & 0 \\
0 & 0 & 0.2 & 0 & 0 & 0 & 0.8 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
\end{array}, \quad
I = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}, \quad
C_\pi = \begin{bmatrix} 0.5 \\ 0 \\ 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}
$$

Using Python with the numpy library we obtained the following:

$$
J^\pi = \begin{bmatrix} 5 & 4.390 & 5 & 5 & 5 & 5 & 5 \end{bmatrix}^\top .
$$